

UNIT-I ELECTRO CHEMISTRY

INTRODUCTION:

Electro chemistry is the branch of physical chemistry. That is concerned with the inter relationship between electricity and chemistry.

Electro chemistry deals with the interaction between electrical energy and chemical species.

These interactions are mainly two types.

Conversion of electrical energy into chemical energy.

Conversion of chemical energy into electrical energy.

TERMINOLOGY:

ELECTRICAL CONDUCTOR: The substance which allows the passage of an electric current through it is called electrical conductor. Ex: all metals (Fe, Cu, Ag), aqueous solutions of acids, bases etc.

ELECTRICAL INSULATOR: The substance which does not allows the passage of an electric current through it is called electrical insulator (or) non conductor. Ex: wood, plastic, paper, rubber, glass etc.

ELECTROLYTE: The substance which is in molten (fusion) or solution state have the capacity to the passage of current through it simultaneously undergoes chemical transformation is called electrolyte.

STRONG ELECTROLYTE: The electrolyte which ionizes completely on dilution is called strong electrolyte.

Ex: strong acids (HCl, H_2SO_4), strong bases (NaOH), salt solutions (NaCl, KCl)

WEAK ELECTROLYTE: The electrolyte which ionizes partially on dilution is called weak electrolyte.

Ex: weak acids (CH_3COOH), weak bases (NH_4OH), salt solution (CH_3COONa)

ELECTROLYSIS: The decomposition of an electrolyte by passing electric current through its solution is called electrolysis.

Electrical conductors are classified into two types.

1. Metallic conductors
2. Electrolytic conductors

1. METALLIC CONDUCTORS	2. ELECTROLYTIC CONDUCTORS
i. These are the metals which conduct electricity. ii. By the passage of electric current no chemical reaction takes place. iii. Free electrons are responsible for electrical conduction. iv. Metallic conduction decreases with increasing temperature (since resistance increases).	i. The substance which is in aqueous solution (or) molten state and liberates ions. ii. By the passage of electric current some chemical reactions take place. iii. Free ions are responsible for electrical conduction. iv. Electrolytic conduction increases with increasing temperature. Due to the increasing the no. of free ions.

Arrhenius Theory:

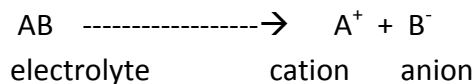
It is applicable only for electrolytic conductors.

1. In solutions, all electrolytes are spontaneously dissociated into charged particles called ions.

The ions carrying +Ve charge are called cations. While the ions carrying -Ve charge are called anions.

Electrolytic dissociation (or) ionization:

The process of spontaneous splitting up of an electrolyte into ions, when dissolved in a solvent is called electrolytic dissociation (or) ionization.



2. The electrolyte at a given dilution may not be completely ionized and the fraction of the total molecules ionized is termed degree of ionization (α).

Degree of ionization = no. of molecules dissociated into ions / total no. of molecules taken.

3. Strong electrolytes are more completely ionized.

Ex: $\text{HCl} \xrightarrow{\hspace{1cm}} \text{H}^+ + \text{Cl}^-$

$\text{NaCl} \xrightarrow{\hspace{1cm}} \text{Na}^+ + \text{Cl}^-$

$\text{KOH} \xrightarrow{\hspace{1cm}} \text{K}^+ + \text{OH}^-$

Degree of ionization 1 (or) 100 % for strong electrolytes.

4. Weak electrolytes are split into ions in small fraction.

Ex: $\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$

$\text{CH}_3\text{COOH} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}^+$

Degree of ionization is less than 1 for weak electrolytes.

5. The conduction of electrolyte increases with increasing temperature.

CONDUCTANCE:

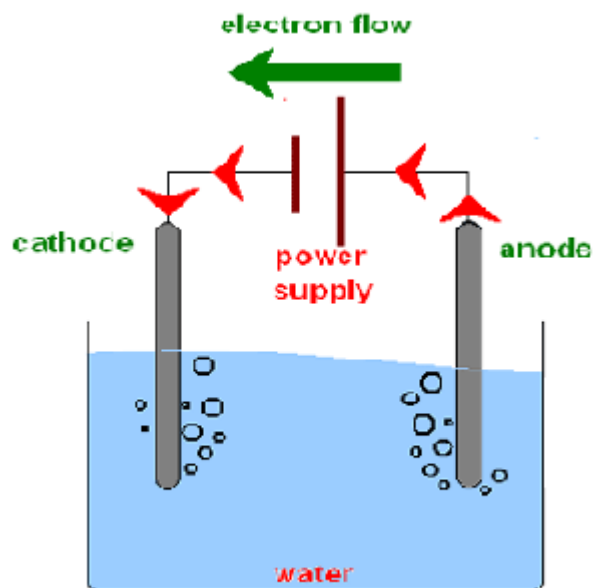
The conducting power of all the ions present in the solution is called conductance. It is denoted by 'C'.

$$C = \frac{1}{R}$$

UNITS: ohm^{-1} (or) Siemens (S) (or) mho

OHM'S LAW:

The resistance of a conductor is directly proportional to the 'l' and inversely proportional to its area of cross section(a).



$l \rightarrow$ distance between the electrodes

$a \rightarrow$ area of cross section

$$R \propto l \text{-----(I)}$$

$$R \propto \frac{1}{a} \text{-----(II)}$$

From I,II

$$R \propto \frac{l}{a}$$

$$R = \rho \frac{l}{a}$$

$$\frac{1}{\rho} = \frac{1}{R} \times \frac{l}{a}$$

$$K = c \times \frac{l}{a}$$

For particular cell, l & a values are constant.

$$K = c \times \text{cell constant}$$

$$K = \frac{1}{R} \times \text{cell constant}$$

UNITS:

$$R = \text{ohm } (\Omega)$$

$$C = \text{ohm}^{-1} (\Omega^{-1}) \text{ (or) Siemens (S)}$$

$$\rho = \text{ohm cm (or) ohm m}$$

$$k = \text{ohm}^{-1} \text{ cm}^{-1} \text{ (or) ohm}^{-1} \text{ m}^{-1}$$

$$\text{cell constant } (l/a) = \text{cm}^{-1} \text{ (or) m}^{-1}$$

Effect of dilution: Up on dilution conductance increases.

SPECIFIC CONDUCTANCE(K): The conducting power of all the ions present in one unit volume (1 ml (or) 1 cc) of the solution is called specific conductance. It is denoted by 'k'. it is independent of resistance, cell constant (l/a) and conductance.

EFFECT OF DILUTION: upon dilution specific conductance decreases. Because on dilution the no. of current carrying ions present per 1 cc of the solution becomes less. Hence specific conductance decreases.

$$\text{UNITS: ohm}^{-1} \text{ cm}^{-1}$$

EQUIVALENT CONDUCTANCE(Λ_v):

The conducting power of all the ions of one gram equivalent electrolyte present in the solution is called equivalent conductance. It is denoted by Λ_v .

EFFECT OF DILUTION: Upon dilution equivalent conductance of the solution increases, due to the increasing no. of free ions present in one gram equivalent of the solution.

RELATION SHIP BETWEEN EQUIVALENT CONDUCTANCE & SPECIFIC CONDUCTANCE:**CASE – I:**

Let us assume one gm equivalent of an electrolyte is present in 1 cc of solution,

Then,

$$\Lambda = k$$

CASE –II:

Let us assume one gm equivalent of an electrolyte is present in 'V' cc of solution, then

$$\Lambda = k \times V$$

Where Λ = equivalent conductance

K= specific conductance

V= volume of the solution

RELATION SHIP BETWEEN EQUIVALENT CONDUCTANCE & NORMALITY:**CASE –I**

If 'k' is expressed on $\text{ohm}^{-1} \text{cm}^{-1}$

Let us assume the normality of the solution 'N'.

Therefore 'N' gram equivalents (GEN) of an electrolyte is present in 1L = 1000 cc of solution.

Then 1 GEN of electrolyte is present in $1000/N \text{ cm}^3$

$$\Lambda = K \times \frac{1000}{N}$$

CASE – II:

If 'k' is expressed in $\text{ohm}^{-1}\text{m}^{-1}$

'N' GEN of an electrolyte is present in $1\text{L} = 10^{-3} \text{ m}^3$

$1 \text{ L} = 10^3 = 10^3(10^{-2} = 10^{-3} \text{ m}^3)$

GEN of an electrolyte is present in $\frac{10^{-3}}{N} \text{ m}^3$.

$$\Lambda = K \times \frac{10^{-3}}{N}$$

UNITS:

$$\begin{aligned}\Lambda &= K \times \frac{1000}{N} \\ &= \frac{\text{ohm}^{-1}\text{cm}^{-1}}{\text{gmequ} / \text{lit}} \\ &= \frac{\text{ohm}^{-1}\text{cm}^{-1}}{\text{gmequ} / \text{cm}^3} = \text{ohm}^{-1}\text{cm}^2\text{gm equ}^{-1}\end{aligned}$$

$$= \text{ohm}^{-1}\text{cm}^2\text{gm equ}^{-1}$$

MOLAR CONDUCTANCE(M):

The conducting power of all the ions of one mole of an electrolyte present in the solution is called molar conductance. It is denoted by μ .

EFFECT OF DILUTION: Upon dilution molar conductance of the solution increases, due to the increasing no. of free ions present in one mole of the solution.

RELATION SHIP BETWEEN MOLAR CONDUCTANCE & SPECIFIC CONDUCTANCE:

CASE – I:

Let us assume one mole of an electrolyte is present in 1 cc of solution,

Then,

$$\mu = k$$

CASE –II:

Let us assume one mole of an electrolyte is present in 'V' cc of solution, then

$$\mu = k \times V$$

Where μ = molar conductance

K= specific conductance

V= volume of the solution

RELATION SHIP BETWEEN MOLAR CONDUCTANCE & MOLARITY:

CASE –I

If 'k' is expressed on ohm-1 cm-1

Let us assume the molarity of the solution 'M'.

Therefore 'M' no.moles of an electrolyte is present in 1L = 1000 cc of solution.

Then 1 mole of an electrolyte is present in $1000/M \text{ cm}^3$

$$\mu = K \times \frac{1000}{M}$$

CASE – II:

If 'k' is expressed in ohm⁻¹m⁻¹

'N' GEN of an electrolyte is present in 1L = 10^{-3} m^3 (1 L = $10^3 \text{ cm}^3 = 10^3(10^{-2} \text{ m})^3 = 10^{-3} \text{ m}^3$)

1 mole of an electrolyte is present in $\frac{10^{-3}}{M} \text{ m}^3$.

$$\Lambda = K \times \frac{10^{-3}}{M}$$

UNITS:

$$\begin{aligned}\Lambda &= K \times \frac{1000}{M} \\ &= \frac{\text{ohm}^{-1} \text{cm}^{-1}}{\text{mole} / \text{lit}} \\ &= \frac{\text{ohm}^{-1} \text{cm}^{-1}}{\text{mole} / \text{cm}^3} = \text{ohm}^{-1} \text{cm}^2 \text{mol}^{-1} \\ &= \text{ohm}^{-1} \text{cm}^2 \text{mol}^{-1}\end{aligned}$$

RELATION SHIP BETWEEN EQUIVALENT CONDUCTANCE & MOLAR CONDUCTANCE:

$$\Lambda = K \times \frac{1000}{N} \text{ -----(I)}$$

$$\mu = K \times \frac{1000}{M} \text{ ----- (II)}$$

$$\text{then I/II} \Rightarrow \frac{\Lambda}{\mu} = \frac{\frac{k \times 1000}{N}}{\frac{k \times 1000}{M}}$$

$$\frac{\Lambda}{\mu} = \frac{M}{N}$$

$$\boxed{\frac{\Lambda}{\mu} = \frac{1}{Z}}$$

Where Z is charge.

UNITS:

Resistance (R) = ohm (Ω)

Conductance (C) = ohm⁻¹ (Ω^{-1}) (or) Siemens (S)

Specific resistanc(ρ)= ohm cm (or) ohm m

Specific conductance (k) = ohm⁻¹ cm⁻¹(or) ohm⁻¹ m⁻¹

cell constant (l/a) = cm⁻¹ (or) m⁻¹

equivalent conductance = = ohm⁻¹ cm²gm equ⁻¹

molar conductance: = ohm⁻¹ cm²mol⁻¹.

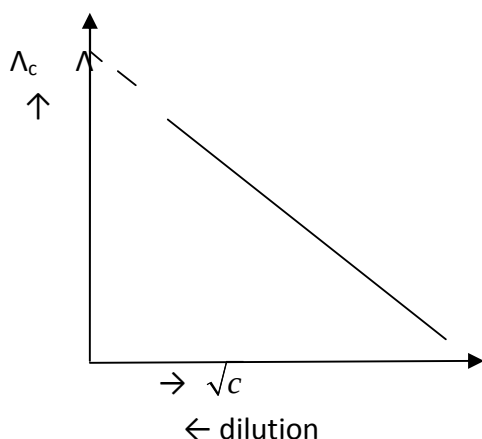
EFFECT OF DILUTION:

conductance mainly depends on three factors.

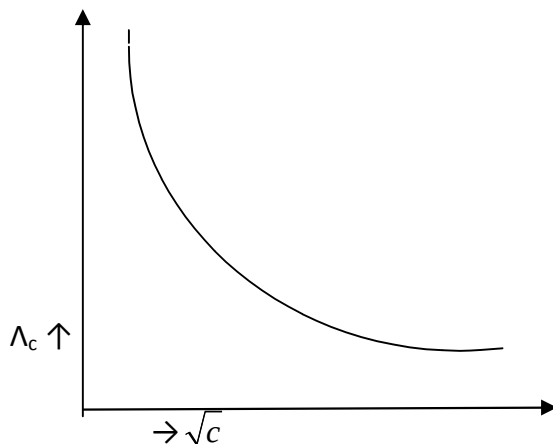
- No. of ions
- The charge of ions
- Mobility of ions.

The conductance increases if we increase the concentration of electrolyte to a certain maximum level and any further increase results in a fall in the conductance value, because till the saturation limit of an electrolyte, increases the population of ions and then reasonable increase in the conductance observed. But the no. of ions an electrolyte are decided by the ionic character of a bond. Further increase in concentration of a solute, the ions get closer due to an increase in the electrostatic force of attraction and viscosity of the medium. These factors decrease the conductance of the solution. Upon dilution for both strong & weak electrolytes equivalent conductance increases.

Λ_c VS $\sqrt{c}$ CURVE FOR STRONG ELECTROLYTES:



Λ_c VS $\sqrt{c}$ CURVE FOR WEAK ELECTROLYTES:



For strong electrolytes, the value of Λ^0 (or) Λ^∞ can be calculated by the extrapolation of Λ_c Vs $\sqrt{c}$ curve .

For weak electrolytes the value of Λ^0 (or) Λ^∞ can not be calculated by the extrapolation of Λ_c Vs $\sqrt{c}$ curve .

For weak electrolytes, the value of Λ^0 (or) Λ^∞ can be calculated indirectly by using Kohlrausch's law.

KOHLRAUSCH'S LAW OF INDEPENDENT MIGRATION OF IONS:

At infinite dilution (where 100 % dissociation takes place) each ionic species (cation (or) anion) contributes a fixed value towards equivalent conductance of an electrolyte irrespective of the other ion in combination with it.

Equivalent conductance at infinite dilution = sum of ionic conductances of the contributing ions

That means $AB \rightarrow A^+ + B^-$

$$\Lambda_{AB}^\infty = \lambda_A^\infty + \lambda_B^\infty$$

Ex: 1. $\Lambda_{HCl}^\infty = \lambda_H^\infty + \lambda_{Cl}^\infty$

2. $\Lambda_{NH_4OH}^\infty = \lambda_{NH_4}^\infty + \lambda_{OH}^\infty$

3. $\Lambda_{HNO_3}^\infty = \lambda_H^\infty + \lambda_{NO_3}^\infty$

APPLICATIONS OF KOHLRAUSCH'S LAW:

1. CALCULATION OF EQUIVALENT CONDUCTIVITY AT INFINITE DILUTION (Λ^0) FOR WEAK ELECTROLYTES.

The extrapolation technique is used for the determination of Λ^∞ of strong electrolytes, it is not useful for the determination of Λ^∞ of weak electrolytes.

Ex 1: CH_3COOH

$$\Lambda_{(CH_3COOH)}^\infty = \lambda_{CH_3COO^-}^\infty + \lambda_{H^+}^\infty$$

By adding & subtracting the Λ^∞ of the salt of strong acid and strong base like HCl & NaOH. ($HCl + NaOH \rightarrow NaCl$)

$$\Lambda_{CH_3COOH}^\infty = \lambda_{CH_3COO^-}^\infty + \lambda_{H^+}^\infty$$

$$= \lambda_{CH_3COO^-}^\infty + \lambda_{H^+}^\infty + \Lambda_{NaCl}^\infty - \Lambda_{NaCl}^\infty$$

$$= [\lambda_{CH_3COO^-}^\infty + \lambda_{H^+}^\infty] + [\lambda_{Na^+}^\infty + \lambda_{Cl^-}^\infty] - [\lambda_{Na^+}^\infty + \lambda_{Cl^-}^\infty]$$

$$\begin{aligned}
&= [\lambda_{\text{CH}_3\text{COO}^-}^\infty + \lambda_{\text{Na}^+}^\infty] + [\lambda_{\text{H}^+}^\infty + \lambda_{\text{Cl}^-}^\infty] - [\lambda_{\text{Na}^+}^\infty + \lambda_{\text{Cl}^-}^\infty] \\
&= \Lambda_{\text{CH}_3\text{COONa}}^\infty + \Lambda_{\text{HCl}}^\infty - \Lambda_{\text{NaCl}}^\infty
\end{aligned}$$

These all are (CH_3COONa , HCl , NaCl) are strong electrolytes. Then Λ^∞ value of these electrolytes can be calculated by using extrapolation technique.

Ex 2: NH_4OH

$$\Lambda_{\text{NH}_4\text{OH}}^\infty = \lambda_{\text{NH}_4^+}^\infty + \lambda_{\text{OH}^-}^\infty$$

By adding & subtracting the Λ^∞ of the salt of strong acid and strong base like HCl & KOH .

($\text{HCl} + \text{KOH} \rightarrow \text{KCl}$)

$$\begin{aligned}
\Lambda_{\text{NH}_4\text{OH}}^\infty &= \lambda_{\text{NH}_4^+}^\infty + \lambda_{\text{OH}^-}^\infty \\
&= \lambda_{\text{NH}_4^+}^\infty + \lambda_{\text{OH}^-}^\infty + \Lambda_{\text{KCl}}^\infty - \Lambda_{\text{KCl}}^\infty \\
&= [\lambda_{\text{NH}_4^+}^\infty + \lambda_{\text{OH}^-}^\infty] + [\lambda_{\text{K}^+}^\infty + \lambda_{\text{Cl}^-}^\infty] - [\lambda_{\text{K}^+}^\infty + \lambda_{\text{Cl}^-}^\infty] \\
&= [\lambda_{\text{NH}_4^+}^\infty + \lambda_{\text{Cl}^-}^\infty] + [\lambda_{\text{K}^+}^\infty + \lambda_{\text{OH}^-}^\infty] - [\lambda_{\text{K}^+}^\infty + \lambda_{\text{Cl}^-}^\infty] \\
&= \Lambda_{\text{NH}_4\text{Cl}}^\infty + \Lambda_{\text{KOH}}^\infty - \Lambda_{\text{KCl}}^\infty
\end{aligned}$$

PROBLEMS:

1. $\Lambda_{\text{BaCl}_2}^\infty = x_1$, $\Lambda_{\text{H}_2\text{SO}_4}^\infty = x_2$, $\Lambda_{\text{HCl}}^\infty = x_3$, $\text{ohm}^{-1} \text{cm}^2 \text{equ}^{-1}$ then calculate the $\Lambda_{\text{BaSO}_4}^\infty = ?$

$$\text{Answer: } \Lambda_{\text{BaSO}_4}^\infty = x_1 + x_2 - x_3 \text{ ohm}^{-1} \text{cm}^2 \text{equ}^{-1}$$

2. $\mu_{\text{BaCl}_2}^\infty = x_1$, $\mu_{\text{H}_2\text{SO}_4}^\infty = x_2$, $\mu_{\text{HCl}}^\infty = x_3$, $\text{ohm}^{-1} \text{cm}^2 \text{equ}^{-1}$ then calculate the $\mu_{\text{BaSO}_4}^\infty = ?$

Answer:

$$\begin{aligned}
\mu_{\text{BaSO}_4}^\infty &= \lambda_{\text{Ba}^{2+}}^\infty + \lambda_{\text{SO}_4^{2-}}^\infty \\
&= [\lambda_{\text{Ba}^{2+}}^\infty + \lambda_{\text{SO}_4^{2-}}^\infty] + 2[\lambda_{\text{H}^+}^\infty + \lambda_{\text{Cl}^-}^\infty] - 2[\lambda_{\text{H}^+}^\infty + \lambda_{\text{Cl}^-}^\infty]
\end{aligned}$$

$$\begin{aligned}
\mu_{\text{BaSO}_4}^\infty &= \mu_{\text{BaCl}_2}^\infty + \mu_{\text{BaSO}_4}^\infty - \mu_{\text{HCl}}^\infty \\
&= x_1 + x_2 - 2x_3
\end{aligned}$$

2. CALCULATION OF DEGREE OF DISSOCIATION & DISSOCIATION CONSTANT OF WEAK ELECTROLYTE:

We know that

$$\Lambda_c \text{ (or) } \Lambda_v \propto \alpha$$

$$\frac{\Lambda_{c_1}}{\Lambda_{c_2}} = \frac{\alpha_1}{\alpha_2}$$

$$\text{If } \alpha_2 \rightarrow 1, \Lambda_{c_2} \rightarrow \Lambda_{\infty}$$

$$\alpha = \frac{\Lambda_{c_1}}{\Lambda_{\infty}}$$

$$\alpha = \frac{\Lambda_c}{\Lambda_{\infty}}$$

the ratio of equivalent conductivity at any dilution to that of infinite dilution is called conductance ratio.

$$\alpha = \frac{\Lambda_c}{\Lambda_{\infty}}$$

For strong electrolytes $\alpha = 1$, but for weak electrolytes ratio is quite small.

Let	CH_3COOH	$\rightleftharpoons$	$\text{CH}_3\text{COO}^- + \text{H}^+$
Initial for 1 mole	1		0 0
at equilibrium	$1(1-\alpha)$		1α 1α
for 'C' moles	$c(1-\alpha)$		$c\alpha$ $c\alpha$

dissociation constant,

$$K_D = \frac{[\text{CH}_3\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{COOH}]} = \frac{[C\alpha][C\alpha]}{c(1-\alpha)}$$

$$K_D = \frac{C\alpha^2}{1-\alpha}$$

where $\alpha = \frac{\Lambda_c}{\Lambda_\infty}$

c = concentration

Λ_c = equivalent conductance at any dilution.

3. DETERMINATION OF SOLUBILITY PRODUCT:

Salts like AgCl, PbSO₄, AgI, Ba SO₄, PbI₂ etc are soluble to a negligible extent and hat ever that gets into solution dissociate completely. If 'S' is the solubility in gm equ lit⁻¹ of a given salt and 'k' is the specific conductance of saturated solution, then Λ_v (or) Λ_v of solution is

$$\Lambda_v = \frac{1000 \times k}{c}$$

These solution are so dilute

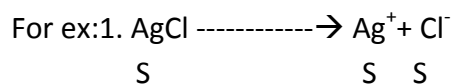
$$\therefore \Lambda_v = \Lambda_\infty$$

$$\Lambda_\infty = \frac{1000 \times k}{S}$$

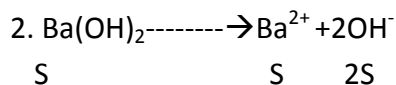
$$S = \frac{1000 \times k}{\Lambda_\infty}$$

For sparingly soluble salts $k_{\text{solution}} = k_{\text{salt}} + k_{\text{water}}$

$$K_{\text{salt}} = k_{\text{solution}} - k_{\text{water}}$$

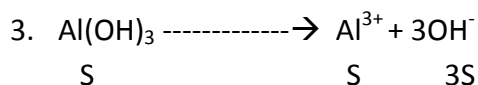


$$\begin{aligned} \text{Solubility product of AgCl} &= [\text{Ag}^+][\text{Cl}^-] \\ &= s.s = S^2 \end{aligned}$$

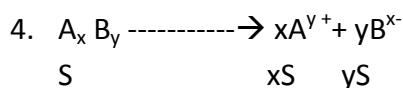


Solubility product of $\text{Ba(OH)}_2 = [\text{Ba}^{2+}][\text{OH}^-]^2$

$$= S.(2S)^2 = 4S^3$$



$$\begin{aligned} \text{S.P} &= S.(3S)^3 \\ &= 27S^4 \end{aligned}$$



$$\begin{aligned} \text{S.P} &= (xs)^x \cdot (ys)^y \\ &= x^x \cdot y^y \cdot s^{(x+y)} \end{aligned}$$

ELECTRO MOTIVE FORCE (EMF):

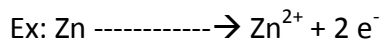
In a galvanic cell the flow of electrons from one electrode to the other electrode takes place, due to this flow of electrons a voltage difference is observed between two electrodes. This voltage difference between two electrodes is called the electro motive force. It is denoted by E. it can be measured by connecting a voltmeter between the two electrodes.

The EMF of a galvanic cell is measured in volts. Then it is called the cell voltage (or) cell potential (E_{cell}).

The EMF is an intensive property and therefore independent of the volume of the solution and weight of the electrodes.

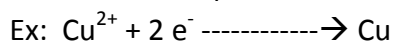
OXIDATION:

Oxidation is a process, which involves the loss of e^- by substance.



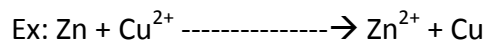
REDUCTION:

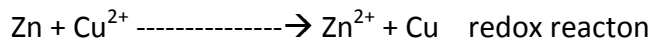
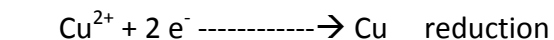
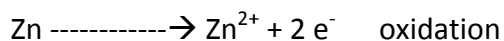
Reduction is a process, which involves the gain of e^- by a substance.



REDOX REACTION:

It is process which involves both oxidation and reduction.





There are two types of cells in which the electro chemical process carried out.

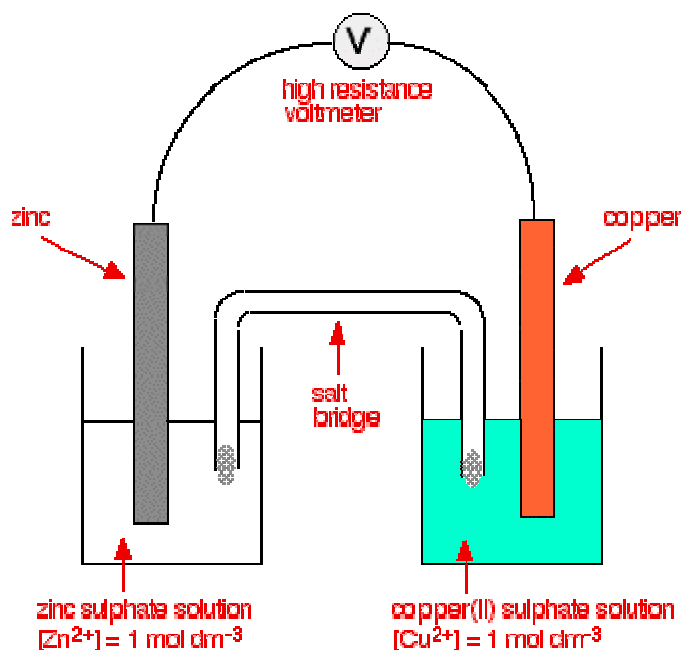
1. ELECTROLYTIC CELL:

The cell in which electrical energy is converted into chemical energy is called electrolytic cell.

2. ELECTRO CHEMICAL CELL:

The cells in which chemical energy is converted into electrical energy is called electro chemical cell.

CONSTRUCTION:



The container A has Zn electrode dipped in aq. ZnSO_4 . The container B has copper electrode dipped in aq. CuSO_4 . The two electrodes are connected by a wire through a voltmeter. An inverted U shape glass tube containing KCl solution in gelatin connects the two containers, it is called salt bridge. The openings of U tube are loosely plugged with cotton to prevent the KCl solution from dipping into the container. It allows the cations and anions to move across. As the redox reaction progresses e^{-} s flow s from anode (Zn electrode) to cathode (Cu electrode).

Reaction at anode: $\text{Zn} \longrightarrow \text{Zn}^{2+} + 2\text{e}^-$

Reaction at cathode: $\text{Cu}^{2+} + 2\text{e}^- \longrightarrow \text{Cu}$

Net reaction: $\text{Zn} + \text{Cu}^{2+} \longrightarrow \text{Zn}^{2+} + \text{Cu}$

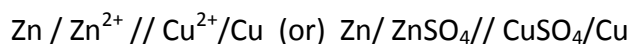
SALT BRIDGE: it is inverted U – shaped glass tube filled with paste of KCl (or) KNO_3 (or) NH_4NO_3 with a gum agar – agar.

USES OF SALT BRIDGE:

It completes the circuit and allows the passage of electric current.

It maintains electrical neutrality in both the half cells.

REPRESENTATION OF GALVANIC CELLS:



Galvanic cell represent as above,

Anode is written on the left hand side, while cathode is written on the right hand side.

The anode of the cell is written on the L.H.S. In this case metal is represented first and then electrolyte is represented. These two are separated by a vertical line or semicolon.

Ex: $\text{Zn} / \text{Zn}^{2+}$ (or) $\text{Zn}; \text{Zn}^{2+}$ (or) $\text{Zn} / \text{ZnSO}_4$ (or) $\text{Zn}; \text{ZnSO}_4$

The cathode of the cell is written on the R.H.S. In this case electrolyte is represented first and then metal is represented. These two are separated by a vertical line or semicolon.

Ex: $\text{Cu}^{2+} / \text{Cu}$ (or) $\text{Cu}^{2+}; \text{Cu}$ (or) $\text{CuSO}_4 / \text{Cu}$ (or) $\text{CuSO}_4; \text{Cu}$

DIFFERENCES BETWEEN ELECTROLYTIC AND ELECTRO CHEMICAL CELL:

Electrolytic cell	Electro chemical cell
1. In these cells electrical energy is Converted into chemical energy.	1. In these cells chemical energy is Converted into electrical energy.
2. Anode is +Ve electrode, cathode is –Ve electrode.	2. Anode is -Ve electrode, cathode +Ve electrode.
3. Anode and cathode are present in the same vessel.	3. Anode and cathode are present in the different vessels.
4. During working of the cell Concentration of electrolyte decreases.	4. During working of the cell Concentration of electrolyte Increases in anode half cell & Concentration of electrolyte decreases In cathode half cell.
5. Salt bridge (or) porous plug is not Required.	5. Salt bridge (or) porous plug is required.

SIMILARITIES:

In both the half cells oxidation takes place at anode and reduction takes place at cathode.

In both the cells e^- flows from anode to cathode in the external circuit.

PROBLEMS:

1. In a conductivity cell 0.01 N KCl solution gave a resistance of 225 ohms, while a 0.01N solution of HCl gave a resistance of 77.1 ohms. Specific conductance of 0.01 N KCl is $0.0014 \text{ ohm}^{-1}\text{cm}^{-1}$.

SOLUTION:

$K = \text{cell constant} \times \text{cell constant}$

$$l/a = k \times 1/c$$

$$= k \times R$$

$$= 0.00141 \times 225 = 0.31725.$$

$$K_{\text{HCl}} = \frac{0.31725}{77.1} = 0.0041 \text{ ohm}^{-1}\text{cm}^{-1}$$

$$\Lambda_{\text{HCl}} = \frac{1000 \times K}{N} = \frac{1000 \times 0.0041}{0.01}$$

$$= 410 \text{ ohm}^{-1}\text{cm}^{-1}$$

2. At ' ∞ ' dilution the equivalent conductance's of NaCl, NaOH, NH_4Cl are $108.9 \text{ Scm}^2 \text{ equ}^{-1}$, $217.4 \text{ Scm}^2 \text{ equ}^{-1}$, $129.8 \text{ Scm}^2 \text{ equ}^{-1}$ respectively. If the equivalent conductivity of N/100 solution of NH_4OH is $9.33 \text{ Scm}^2 \text{ equ}^{-1}$, calculate the degree of dissociation of NH_4OH .

Solution:

$$\Lambda_{\text{NH}_4\text{Cl}}^{\infty} = \lambda_{\text{NH}_4^+}^{\infty} + \lambda_{\text{Cl}^-}^{\infty}$$

$$\Lambda_{\text{NH}_4\text{OH}}^{\infty} = \Lambda_{\text{NH}_4\text{Cl}}^{\infty} + \Lambda_{\text{NaOH}}^{\infty} - \Lambda_{\text{NaCl}}^{\infty}$$

$$= 129.8 + 217.4 - 108.9$$

$$\text{Degree of dissociation} = \alpha = \frac{\Lambda_c}{\Lambda_{\infty}} = \frac{9.33}{238.3}$$

$$= 0.039$$

$$\text{Dissociation constant, } K_D = \frac{C\alpha^2}{1-\alpha}$$

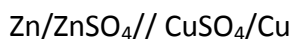
$$= 1.58 \times 10^{-5}$$

SINGLE ELECTRODE POTENTIAL:

Each electro chemical cell is made up of two electrodes, at one electrode e^- s evolved and at another electrode e^- s used up.

Each electrode which is dipped in its salt solution is called half cell.

The potential of half cell i.e. the potential difference between the metal and its salt solution in which it is dipped is called single electrode potential.

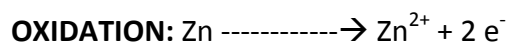


The total cell emf is equal the sum of the single electrode potentials. Each electrode is with a symbol corresponding to the reaction that takes place near the electrode.

$$E_{\text{cell}}^0 = E_{\text{anode}}^0 + E_{\text{cathode}}^0$$

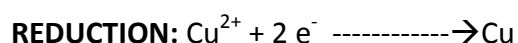
The half cell reactions as follows.

The half cell reaction which corresponds to oxidation,



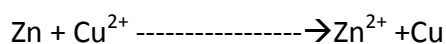
The electrode is called oxidation electrode and the potential of that electrode is oxidation potential (or) the potential of left hand electrode, which represented as E_{ox} or E_{L} .

The half cell reaction which corresponds to reduction,



The electrode is called reduction electrode and the potential of that electrode reduction potential (or) the potential of right hand electrode, which represented as E_{red} or E_{R} .

The total cell reaction is,

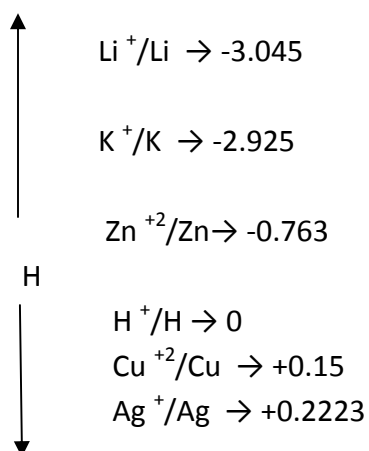


$$E_{\text{cell}} = E_{\text{ox}} + E_{\text{red}}$$

Emf of total cell is equal to the sum of oxidation potential & reduction potential.

GALVANIC SERIES (OR) ELECTRO CHEMICAL SERIES:

When the metals are arranged in the order of increasing reduction potentials (or) decreasing oxidation potentials which are determined with respect to one molar solutions of their ions and measured on the hydrogen scale, along series is called electro chemical (or) galvanic series.



The higher a metal in the series, the greater is tendency to be oxidized. The metals high up in the series are strong reducing agents, and their ions are stable than metals.

Where as those metals near the bottom of the series are inactive & stable metals, their ions are easily reduced to metals.

Any metal can reduce any cation below to it.

Ex: $\text{Zn}, \text{Cu}^{2+}$ then Zn reduces the Cu^{2+} to Cu.

In the series any metal above hydrogen will displace hydrogen from dilute acid. for ex: Na reacts with water to liberate hydrogen, because the $E^\circ_{(\text{Na}^+/\text{Na})} = -2.714 \text{ V}$ is less than $E^\circ_{\text{H}^+/\text{H}_2} = 0$.

Any metal below the hydrogen in the series will be reduced by hydrogen.

The metals above the hydrogen in the series can easily displace the metal below the hydrogen in the series from its salt solution. Because $E^\circ_{(\text{Zn}^{2+}/\text{Zn})} = -0.763 \text{ V}$ is less than $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = 0.337 \text{ V}$.

for ex: $\text{Zn} + \text{CuSO}_4 \longrightarrow \text{ZnSO}_4 + \text{Cu}$

The metals above the hydrogen in the series can be easily oxidized, hence they undergoes corrosion easily.

APPLICATIONS OF ELECTRO CHEMICAL SERIES:

- The relative corrosion tendencies of the metals and alloys.
- Relative ease of oxidation (or) reduction of metals.
- Replacement tendency of metals.
- Predicting spontaneity of redox reactions.
- Calculating the equilibrium constant as given below.

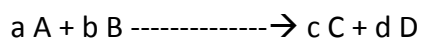
$$\log K_{\text{equ}} = \frac{2.303 E^\circ}{RT/nF}$$

$$\log K_{\text{equ}} = \frac{E^\circ}{2.303/n} \text{ at } 25^\circ\text{C}$$

$$R = 8.314 \text{ J mol}^{-1}\text{K}^{-1}$$

NERNST EQUATION:

Nernst gave the relationship between the electrochemical reaction and the corresponding cell emf. This relationship is generally known as Nernst equation.



Where a, b, c, d are corresponding to no. of moles of A, B, C, D respectively at equilibrium.

$$E_{\text{cell}} = \frac{RT}{nF} \ln k_{\text{equ}} - \frac{RT}{nF} \ln \frac{a_{\text{products}}}{a_{\text{reactants}}}$$

For solids $a=1$, for liquids $a=1$

The Nernst of cell can be written as,

$$E_{cell} = \frac{RT}{nF} \ln k_{equ} - \frac{RT}{nF} \ln \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

Here

E_{cell} is the cell potential ([electromotive force](#))

$E^{\ominus}_{cell}$ is the *standard* cell potential at the temperature of interest

R is the [universal gas constant](#): $R = 8.314 \text{ J K}^{-1}$

T is the [absolute temperature](#)

F is the [Faraday constant](#), $F = 96457 \text{ C}$

n is the number of moles of [electrons](#) transferred in the cell reaction or [half-reaction](#).

Quantities in square brackets refers the respective concentrations .

The quantity $\left(\frac{RT}{nF} \right) \ln k$ is known as standard EMF of the cell an $E^{\ominus}_{cell}$.

Then the equation is
$$E_{cell} = E^{\ominus}_{cell} \ln k_{equ} - \frac{RT}{nF} \ln \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

n is the number of moles of [electrons](#) transferred in the cell reaction or [half-reaction](#) at roomtemperature ($T=298\text{K}$), taking $R=8.314\text{J K}^{-1}$, $F=96497\text{C}$.

PROBLEMS:

3. In a conductivity cell 0.01 N KCl solution gave a resistance of 225 ohms, while a 0.01N solution of HCl gave a resistance of 77.1 ohms. Specific conductance of 0.01 N KCl is $0.0014 \text{ ohm}^{-1}\text{cm}^{-1}$.

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$$\Lambda_{\text{NH}_4\text{OH}}^{\infty} = \Lambda_{\text{NH}_4\text{Cl}}^{\infty} + \Lambda_{\text{NaOH}}^{\infty} - \Lambda_{\text{NaCl}}^{\infty}$$

$$= 129.8 + 217.4 - 108.9$$

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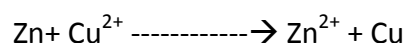
$$= 0.039$$

$$\text{Dissociation constant, } K_D = \frac{C\alpha^2}{1-\alpha}$$

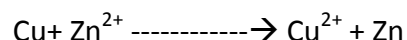
$$= 1.58 \times 10^{-5}$$

REVERSIBLE AND IRREVERSIBLE CELLS:

The emf of the Daniel cell is 1.09 volt and the cell reaction is,



If to such a cell an external emf which is exactly to 1.09 volt is applied to the cell, reaction stops. This is taken as the basis for the measurement of emf of a cell, but if the external emf is increases infinitesimally by small amount, then cell reaction is reversed as given below.



If the external emf applied is less than 1.09 volt then no change in the cell reaction takes place. This type of cell is called reversible cell. The electrical energy of such reversible cell is calculated by the free energy decrease of the reaction in the cell.

For spontaneous reaction,

$$\Delta G = -nFE$$

ΔG = decrease in free energy

n = valency of ion

E = emf of the cell

F = faraday

*

Reaction at anode: $\text{Zn} \longrightarrow \text{Zn}^{2+} + 2 \text{e}^-$

Reaction at cathode: $\text{Cu}^{2+} + 2 \text{e}^- \longrightarrow \text{Cu}$

spontaneous reaction: $\text{Zn} + \text{Cu}^{2+} \longrightarrow \text{Zn}^{2+} + \text{Cu}$

*

Reaction at anode: $\text{Cu} \longrightarrow \text{Cu}^{2+} + 2 \text{e}^-$

Reaction at cathode: $\text{Zn}^{2+} + 2 \text{e}^- \longrightarrow \text{Zn}$

Non spontaneous reaction: $\text{Zn}^{2+} + \text{Cu} \longrightarrow \text{Zn} + \text{Cu}^{2+}$

Both spontaneous & non spontaneous reactions are reversible to each other.

Any cell which does not satisfy the above condition is irreversible.

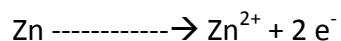
For ex: Zn and Ag electrodes, when the two electrodes are connected zinc ions go into solution.

TYPES OF ELECTRODES:

Important types of electrodes of an electro chemical cell can be grouped into the following.

i. METAL – METAL ION ELECTRODE:

The electrode can be constructed by bringing a zinc rod in contact with zinc ions. Oxidation of Zinc atoms to give Zinc ions is expressed by the following equation.



If the reaction two electrodes are exchanged between both the zinc species in the overall redox reaction making, $n = 2$.

$$\therefore E_{\text{Zn}/\text{Zn}^{2+}} = E_{\text{Zn}/\text{Zn}^{2+}}^0 - \frac{0.0591}{2} \log \frac{a_{\text{Zn}^{2+}}}{a_{\text{Zn}}}$$

Since the activity of all the pure solids is unity ($a = 1$).

$$\therefore E_{\text{Zn}/\text{Zn}^{2+}} = E_{\text{Zn}/\text{Zn}^{2+}}^0 - \frac{0.0591}{2} \log [\text{Zn}^{2+}]$$

The expression suggests that the oxidation electrode potential (E) decreases with increase $[\text{Zn}^{2+}]$ and equals to standard electrode potential E^0 at unit concentration of the electrode.

ii. METAL – METAL INSOLUBLE SALT ELECTRODE:

If consists of a metal (M) covered by layer of sparingly soluble salt (MX) immersed in a solution containing a common ion (X^{-}).

It is represented as, $X^{-}(\text{aq})/\text{MX}/M(\text{S})$

Electrode reaction for the type of electrode can be written as



And the electrode potential

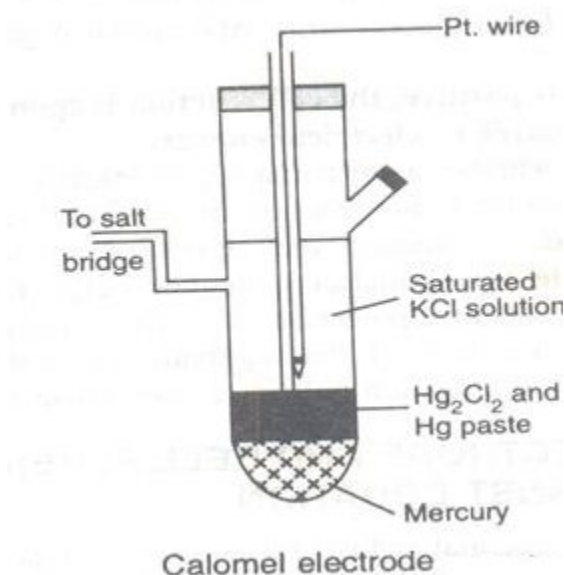
$$E_{X^-/MX/M} = E_{X^-/MX/M}^0 - \frac{RT}{F} \log a_{X^-}$$

ex: calomel electrode.

CALOMEL ELECTRODE (SECONDARY REFERENCE ELECTRODE):

This electrode has Hg in contact with its insoluble salt Hg_2Cl_2 and chloride ions. The second type of reference electrode employed for measuring the single electrode potential is calomel electrode and this electrode is only used as reducing electrode.

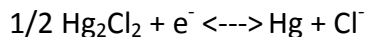
CONSTRUCTION:



1. It is secondary standard electrode consists of glass tube having side tubes on each side.
2. The mercury of high purity is placed at the bottom of this tube and connected to the circuit by means of platinum wire sealed in a glass tube.
3. The surface of mercury is covered with a paste of mercurous chloride.
4. The electrode is connected with the help of a side tube on the left through salt bridge with the other electrode whose potential has to be measured.
5. The electrode represented as $Hg, Hg_2Cl_2/KCl$ (salt solution)

WORKING:

The electrode reaction when the cell acts as a cathode is



And its potential is given by

$$E = E^0_{\text{Cl}^-/\text{Hg}_2\text{Cl}_2/\text{Pt}} - \frac{RT}{F} \log a_{\text{Cl}^-}$$

Thus, the potential of the calomel electrode depend on the activity of the chloride ions (a_{Cl^-}) and increases as the activity of chloride ions decreases. It has been found that the potential of calomel electrode vary with the concentration of the potassium chloride solution is used.

The standard reduction potential of this electrode at 25°C

For Saturated KCl solution; 0.2415 V

For 1N KCl solution: $E^0 = 0.28 \text{ V}$

Hg, $\text{Hg}_2\text{Cl}_2 (\text{S})/\text{Hg}, \text{Pt}$

1.0N KCl 0.2800 V

1M KCl, $\text{Hg}_2\text{Cl}_2 (\text{S})/\text{Hg}, \text{Pt}$

0.1N KCl 0.3338 V

0.1 M KCl, $\text{Hg}_2\text{Cl}_2 (\text{S})/\text{Hg}, \text{Pt}$

ADVANTAGES:

- i. It is simple to construct.
- ii. A result of cell potential measurements is reproducible and stable over a long period and does not vary with temperature.

iii. METAL – AMALGAM ELECTRODE:

It places the metal – amalgam in a contact with a solution of metal ion.

It is more convenient to use the highly reactive metals in the form of amalgams. Since the activity of the metal is lowered by dilution with mercury.

It is represented as: $\text{M}^{n+} (\text{aq}) / \text{M} (\text{Hg})$

Electrode reaction: $\text{M}^{n+} + n \text{e}^- \rightleftharpoons \text{M} (\text{Hg})$

And electrode potential: $E_{M^{n+}} = E_{M^{n+}/M(Hg)}^0 - \frac{RT}{nF} \ln \left(\frac{a_{M(Hg)}}{a_{M^{n+}}} \right) a_{Cl^-}$

It is more convenient to use the highly reactive metals in the form of amalgams. Since the activity of the metal is lowered by dilution with mercury.

It is represented as $M^{n+}/M(Hg)$

Electrode reaction : $M^{n+} (aq) + ne^- \rightleftharpoons M (Hg)$

And electrode potential: $E_{M^{n+}/M} = E_{M^{n+}/M}^0 - \frac{RT}{nF} \ln \left(\frac{a_M}{a_{M^{n+}}} \right)$

iv. GAS – ION ELECTRODE:

It consists of an inert metal (ex: Pt) dipped in a solution conditioning ions to which the gas is reversible. The latter is continuously bubbled through the solution.

It is represented as $X^-(aq) / X_2 (P = X \text{ atm}) / Pt$

Electrode reaction: $X_2 (P) + 2 e^- \rightleftharpoons 2X^- (aq)$

And electrode potential: $E = E_{X^-/X_2/Pt}^0 - \frac{RT}{nF} \ln \left(\frac{a_{X^-}^2}{a_{X_2}} \right)$

Ex: standard hydrogen electrode

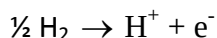
STANDARD HYDROGEN ELECTRODE (PRIMARY REFERENCE ELECTRODE):

The reference electrode or standard Hydrogen half cell (or) standard hydrogen electrode is selected for coupling the half cell.

Construction:

- It consists of platinum wire coated with platinum immersed in a 1M concentration solution of H^+ ions maintained at $25^\circ C$.
- H_2 gas at 1 atm pressure enters the glass hood and bubbles over the platinum electrode.
- The H_2 gas at the platinum electrode passes into the solution forming H^+ ions of unit concentration (1M) is arbitrarily fixed as zero.

- Depending on the half cell to which it is attached hydrogen electrode can acts as cathode or as anode.
- The electrode reaction of this electrode is



By applying the Nernst equation

$$E_{\text{H}_2/\text{H}^+} = E^0_{\text{H}_2/\text{H}^+} - \frac{0.0591}{1} \log \left(\frac{a_{\text{H}^+}}{a_{\text{H}_2}^{1/2}} \right)$$

$$E_{\text{H}_2/\text{H}^+} = E^0_{\text{H}_2/\text{H}^+} - 0.0591 \log(a_{\text{H}^+})$$

$$E_{\text{H}_2/\text{H}^+} = E^0_{\text{H}_2/\text{H}^+} + 0.0591 \log \text{pH} \quad (\because \text{pH} = -\log [\text{H}^+])$$

Advantages:

1. It gives satisfactory results in the entire pH ranges of 0 – 14 .
2. The potential and pH are free salt errors.

Limitations:

1. The hydrogen gas used must be sufficiently pure.
2. The electrode cannot be used in the presence of gas impurities like air & oxygen, electrolyte impurities like oxidizing and reducing agents.
3. It is very susceptible to poisoning by cyanides, Arsenic and Antimony compounds, alkaloids.
4. It is very convenient in handling.

So, due to these drawbacks of hydrogen electrode secondary reference electrode like standard calomel, Quin – Hydrone and glass electrode etc are used.

REDOX ELECTRODE:

It consists of an inert metal (ex: Pt) dipped in a solution containing ions in two oxidation states of the substance (ex: Fe^{2+} and Fe^{3+})

It is represented as $\text{Pt} / \text{M}^{n1+}(a_1), \text{M}^{n2+}(a_2)$

Electrode reaction: $\text{M}^{n1+}(a_1) + n e^- \rightleftharpoons \text{M}^{n2+}$

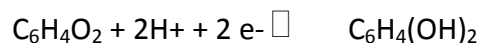
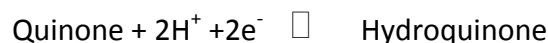
And electrode potential: $E = E^0_{\text{M}^{n1+}, \text{M}^{n2+}} - \frac{RT}{nF} \ln\left(\frac{a_{\text{M}^{n2+}}}{a_{\text{M}^{n1+}}}\right)$

Ex: Quinhydrone electrode.

QUINHYDRONE ELECTRODE:

- It is a redox electrode reversible to protons and often replaces the hydrogen electrode.
- Quinhydrone is 1:1 molar mixture of quinone and hydroquinone.
- The electrode consists of a platinum electrode dipped in an acid/ base test solution, which is saturated with quinhydrone.

The electrode reaction is given by



i.e.

The electrode potential at 25°C is given by

$$E_{\text{Pt}/\text{Q}, \text{H}^+, \text{QH}_2} = E^0_{\text{Pt}/\text{H}^+/\text{QH}_2} - \frac{0.0591}{2} \log\left(\frac{a_{\text{QH}_2}}{a_{\text{Q}} \cdot a_{\text{H}^+}^2}\right)$$

Since quinone and hydroquinone are taken in equimolar amounts.

i.e. $a_{\text{Q}} = a_{\text{QH}_2}$

$$E = E^0 - \frac{0.0591}{2} \log\left(\frac{1}{a_{\text{H}^+}^2}\right)$$

$$E = E^0 - \frac{0.0591}{2} \log(a_{H^+}^{-2})$$

$$E = E^0 + 2 \times \frac{0.0591}{2} \log(a_{H^+})$$

$$E_{Q, QH_2} = E_{Q, QH_2}^0 - 0.0591 pH$$

Quinhydrone electrode is used to measure pH of solution. Due to un stability of quinine in strong alkaline medium. This hydrogen ion indicator electrode is suitable only up to a pH of 8.

ADVANTAGES:

1. The electrode is very easy to setup.
2. The pH value obtained is very accurate.
3. Very small quantities of the solution are sufficient for the measurement.

LIMITATIONS:

1. This electrode cannot be used in more alkaline ($P^H > 8.5$) solutions.
2. This electrode fails in the presence of strong oxidizing & reducing agents.

ION SELECTIVE ELECTRODES:

Ion selective electrodes have the ability to respond to certain specific ions and develop a potential. The potential developed is a mixture of the concentration of the species of interest. These electrodes use a membrane which is sensitive to particular species.

MEASUREMENT OF CONCENTRATION OF M^{n+} IN A SOLUTION:

The cell consisted for the measurement of concentration of cation (M^{n+}) is given below. The membrane potential is given by the equation.

$$E_{M^{n+}} = \frac{RT}{nF} \ln\left(\frac{C_2}{C_1}\right)$$

Suppose in a cell the reference electrode is anode and the membrane electrode is cathode,

$$\therefore \text{The cell potential } E_{cell} = E_{M^{n+}} - \square E_{ref}$$

Here $\square E_{ref} = 0$ hen identical reference electrodes are used.

In ion selective electrodes, the use of membrane is must. Generally the following four types of membranes are used.

1. Glass membrane
2. Solid state membrane (pellets of $\text{Ag}_2\text{S} + \text{Ag}$)
3. Hetero genius membrane.
4. Liquid membrane (tris 1, 10 phenanthralene $\text{Fe}^{2+} (\text{ClO}_4^-)_2$)

Ex: glass electrode (it is the university employed electrode for pH measurement)

GLASS ELECTRODE:

CONTRUCTION:

The glass electrode made of a special glass of relatively low melting point and high electrical conductivity. It is coming glass containing Na_2O (22%), CaO (6%) and SiO_2 (72%)

The assembly of glass electrode consists of a thin glass bulb filled with 0.1N HCl and a silver wire coated with silver chloride (AgCl) is immersed in it. Here Ag/AgCl acts as an internal reference electrode. The glass electrode is represented as $\text{Ag}/\text{AgCl} (\text{S})/0.1\text{m HCl}/\text{glass}$

PRINCIPLE:

When the glass electrode is immersed in another solution, whose pH value is to be determined, there develops a potential between the two surfaces of membrane. The potential difference developed is proportional to the difference in pH value.

THEORY:

The potential difference or the boundary potential (E_B) at the interface is the result of the difference in potential ($E_1 - E_2$) developed across the gel layer of the glass membrane between the two liquids. This boundary potential is related to the concentration of acid solution into which the glass bulb is dipped (C_2) by

$$E_B = E_1 - E_2 = \frac{RT}{nF} \ln\left(\frac{C_2}{C_1}\right)$$

As $C_1 = 0.1 \text{ M}$ (constant)

$$\therefore \text{boundary potential} = (E_B) = \frac{RT}{nF} \ln C_2 - L$$

Where L is a constant and it depends on the pH of the solution taken in the bulb of glass electrode.

$$\text{When } C_1 = C_2 \Rightarrow E_B = 0$$

When $C_1 = C_2$, a small potential is developed. This is called asymmetric potential.

$$E_G = E_B + E_{Ag/AgCl} + E_{ass}$$

$$= \frac{RT}{nF} \ln C_2 + (-L + E_{Ag/AgCl} + E_{ass})$$

$$\therefore \boxed{E_G + E_G^0 - 0.0591 \text{ pH}} \quad (\because E_G^0 = -L + E_{Ag/AgCl} + E_{ass})$$

LIMITATIONS:

- The glass electrode can be used upto a pH of 13. But becomes sensitive to Na^+ ions above $\text{pH} = 9$ resulting in an alkaline error.
- For the measurement of EMF of the glass electrode, we cannot use ordinary potentiometers, because of high resistance of glass. It is therefore, necessary to use special electronic potentiometers.

ADVANTAGES OF GLASS ELECTRODE:

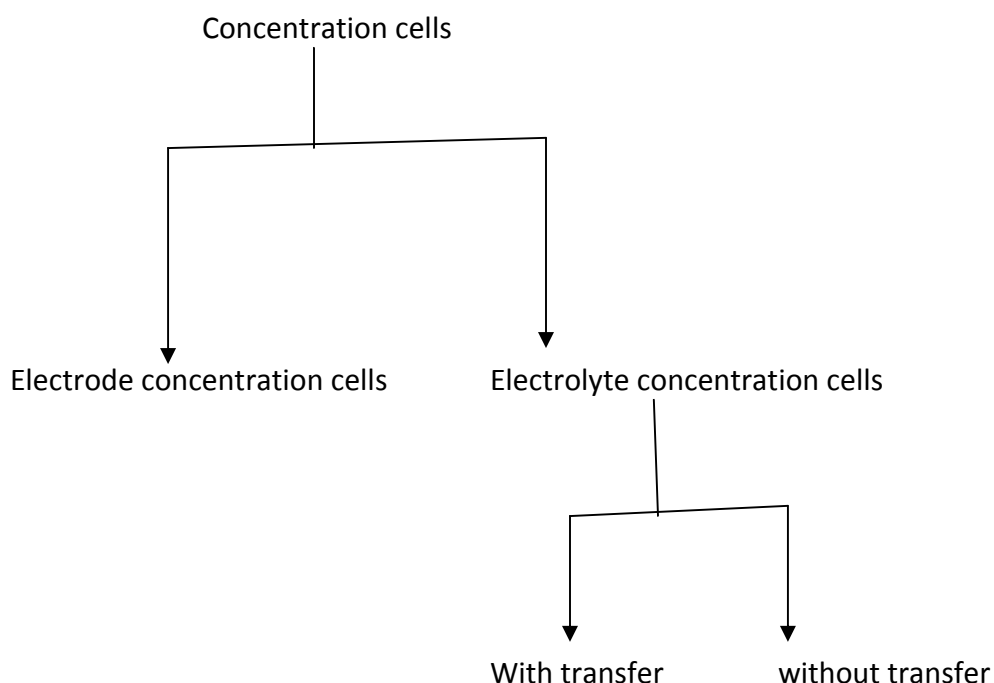
1. It is very convenient and simple to use.
2. It is not easily poisoned.
3. Equilibrium is rapidly achieved.
4. The results are accurate.
5. It has no salt error.
6. It can be used in turbid, colored and colloidal solutions.

7. pH measurement is possible even with few milliliters of solutions.
8. It can be used both in alkaline and oxidizing solutions.

CONCENTRATIONS CELLS:

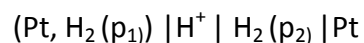
Concentration cell is an electrochemical (or) galvanic cell in which electrical energy produced by the transfer of material from one electrode to another due to difference in concentrations between the two i.e. from a system of high concentration to low concentration. This is contrast to the galvanic cell in which the EMF arises due to decrease in the free energy from a chemical reaction.

In general, there are two types of concentration cells.



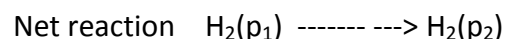
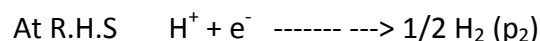
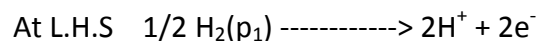
(i) ELECTRODE CONCENTRATION CELLS:

- In these cells, the EMF arises as a result of like electrodes at different concentrations dipping in the same solution of the electrolyte.
- For example, this type of cells consists of two hydrogen electrodes at different gas pressures (p_1 & p_2) immersed in the same solution of hydrogen ions at 25°C represented as
as



❖ If $p_1 > p_2$ oxidation occurs at L.H.S. electrode and reduction occurs at R.H.S. electrode.

The electrode reactions are



The cell reaction involves the transfer of hydrogen gas from pressure (p_1) at the other electrode. In other words the cell reaction is spontaneous. Expansion of H_2 gas from pressure p_1 to p_2 and E_{cell} depends only on two pressures (p_1 & p_2) and is independent on activity of H^+ ions in the solution.

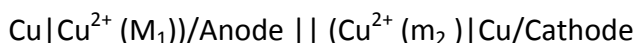
$$\frac{1}{2} \text{H}_2(p_1) = \frac{1}{2} \text{H}_2(p_2)$$

$$E_{\text{cell}} = -0.0591 \log \left(\frac{p_2}{p_1} \right)$$

$$E_{\text{cell}} = 0.0591 \log \left(\frac{p_1}{p_2} \right) \text{ at } 25^\circ \text{C}$$

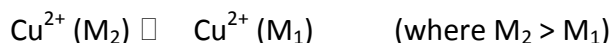
(ii) Electrolyte concentration cells (with transference):

These cells consist of two identical electrodes dipped in two electrolyte solutions of different concentrations. Due to which a difference in potential at the electrode and electrolytic solution in which it is dipped.



Where M_1 and M_2 are the molar concentrations of Cu^{2+} ions in the two half cells.

The net cell reaction is



And at 25°C the cell emf is given by

The emf of the cell is given by the following expression:

$$E_{\text{cell}} = \frac{0.0591}{2} \log \left(\frac{M_2}{M_1} \right)$$

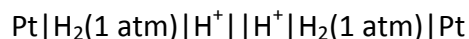
Such a cell, here liquid junctions (solutions of different concentrations are connected through salt bridge) are involved.

Applications:

1. To determine the solubility of sparingly soluble salt.
2. To calculate valency of ions.
3. To determine the extent of corrosion in metals.
4. To determine the equilibrium constant.

Example1. A cell contains two hydrogen electrodes. The negative electrode is in contact with a solution of 10^{-6} M hydrogen ions. The emf of the cell is 0.118 volt at 25° C. Calculate the concentration of hydrogen ions at the positive electrode.

Solution: The cell may be represented as



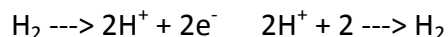
10^{-6} M CM

Anode

Cathode

(-ve)

(+ve)



$$E_{\text{cell}} = 0.0591/2 \log([\text{H}^+]_{\text{cathode}}^2)/[10^{-6}]^2$$

$$0.081 = (0.0591) \log([\text{H}^+])/10^{-6}$$

$$\log[\text{H}^+]_{\text{cathode}}/10^{-6} = 0.118/0.0591 = 2$$

$$[\text{H}^+]_{\text{cathode}}/10^{-6} = 10^2$$

$$[\text{H}^+]_{\text{cathode}} = 10^{-6} \times 10^2 = 10^{-4} \text{ M}$$

Example 2. The emf of the cell $\text{Ag} | \text{AgI} \text{ in } 0.05 \text{ M } \text{NH}_4\text{NO}_3 | 0.05 \text{ M } \text{AgNO}_3 | \text{Ag}$ is 0.788 volt at 25°C . The activity coefficient of KI and silver nitrate in the above solution is 0.90 each. Calculate (i) the solubility product of AgI, and (ii) the solubility of AgI in pure water at 25°C .

Solution: Ag^+ ion concentration on AgNO_3 side

$$= 0.9 \times 0.5 = 0.045 \text{ M}$$

Similarly I^- ion concentration in 0.05 M KI solution

$$= 0.05 \times 0.9 - 0.045 \text{ M}$$

$$E_{\text{cell}} = 0.0591/1 \log [Ag^+]_{(R.H.S.)} / [Ag^+]_{(L.H.S.)} = 0.0591 \log 0.045 / [Ag^+]_{(L.H.S.)}$$

$$\text{or } \log 0.045 / [Ag^+]_{(L.H.S.)} = 0.788 / 0.0591 = 13.33$$

$$[Ag^+]_{L.H.S.} = 0.045 / (2.138 \times 10^{13})$$

$$= 2.105 \times 10^{-15} \text{ M}$$

$$\text{Solubility product of AgI} = [Ag^+][I^-]$$

$$= 2.105 \times 10^{-15} \times 0.045$$

$$= 9.427 \times 10^{-17}$$

$$\text{Solubility of AgI} = \sqrt{\text{Solubility product of AgI}}$$

$$= \sqrt{9.427 \times 10^{-17}}$$

$$= 9.732 \times 10^{-9} \text{ g mol L}^{-1}$$

$$= 9.732 \times 10^{-9} \times 143.5 \text{ g L}^{-1}$$

$$= 1.396 \times 10^{-6} \text{ g L}^{-1}$$

Example 3:

The observed emf of the cell



is 0.154 V. Calculate the value of M_1 and pH of cathodic solution.

$$\text{Solution: } E_{\text{cell}} = 0.0591 \log M_1 / (3 \times 10^{-4})$$

$$\text{or } \log M_1 / (3 \times 10^{-4}) = 0.0154 / 0.0591 = 2.6058$$

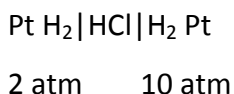
$$M_1 / (3 \times 10^{-4}) = 4.034 \times 10^2$$

$$M_1 = 4.034 \times 10^2 \times 3 \times 10^{-4} \text{ M}$$

$$= 0.121 \text{ M}$$

$$\text{pH} = -\log [H^+] = -\log 0.121 = 0.917$$

Example 4: Calculate the emf of the following cell at 25°C.



Solution: $E_{\text{cell}} = 0.0591/2 \log P_1/P_2$

$$= 0.0591/2 \log 2/10$$

$$= -0.0206 \text{ volt}$$

POTENTIOMETRIC TITRATION:

These are one important application of electrode potentials.

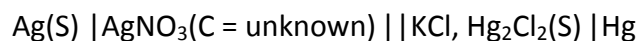
These are defined as:

The titrations in which the equivalent (or) end point of a reaction is determined with the help of the measurement of the potentials of the reaction mixture are known as potentiometric titrations.

1. PRECIPITATION TITRATION:

In precipitation titration an electrode reversible to one of the ions involved is made use of.

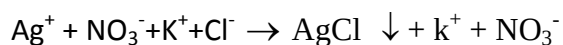
Let us consider the titration of silver nitrate solution of potassium chloride solution. The silver nitrate solution is used to form a reversible electrode. It is combined with a calomel electrode (reference electrode) to form the cell which is presented as:



The electrode potential of the silver electrode depends on the concentration of the silver ions in the solution as

$$E_{\text{Ag}} = E^0_{\text{Ag}} - \frac{RT}{F} \log C_{\text{Ag}^+} \text{ -----} \rightarrow \text{I}$$

During the titration , when small amount of KCl is added, the concentration of silver ions falls due to the occurrence of following reaction.



From equation I

The concentration of Ag^+ is decreased and has low value then the electrode potential of silver electrode E_{Ag} increases much more rapidly than any where else during the titration.

If EMF is plotted against volume of KCl solution added, we notice a sigmoid curve with the step portion indicating the neutralization point.

2. ACID-BASE TITRATIONS:

The acid solution whose strength has to be determined is taken in a beaker and the hydrogen electrode and calomel electrode are dipped in the solution. The electrodes were connected to the potentiometer and the emf is measured. A known volume, standard alkali solution is added from a burette, stirred thoroughly and the emf of the cell recorded like this 10 – 15 readings are recorded by repeating the procedure of the addition of standard alkali.

If we plot graph between emf and volume of NaOH

If we plot graph between $\frac{\Delta E}{\Delta V}$ and volume of NaOH

Cell represented as $\text{Pt, Hg} / \text{Hg}_2\text{Cl}_2 (\text{s}) / \text{KCl (saturated)} // \text{H}^+ / \text{H}_2 (1 \text{ atm}), \text{Pt}$

The emf of indicator electrode is

$$E_{\text{H}_2/\text{H}^+} = E_{\text{H}_2/\text{H}^+}^0 + 0.0591 \log [\text{H}^+] \quad (\because E_{\text{H}_2/\text{H}^+}^0)$$

$$E_{\text{H}_2/\text{H}^+} = 0.0591 \log [\text{H}^+]$$

$$E_{\text{H}_2/\text{H}^+} = -0.0591 \log pH$$

3. OXIDATION – REDUCTION TITRATIONS:

The procedure involved these titrations are similar to the Acid – Base titrations. The only difference is that the electrode reversible to hydrogen ion is replaced by bright platinum electrode. The emf of the electrode is determined by the activity of ratio of the substance being oxidized (or) reduced.

Ex: Fe^{2+} titrated against $\text{K}_2\text{Cr}_2\text{O}_7$, the ferrous ion solution is taken in the beaker, treated with dil. H_2SO_4 and platinum and platinum & calomel electrodes are dipped.

- The electrodes are connected to the potentiometer and emf of the solution after the addition of $\text{K}_2\text{Cr}_2\text{O}_7$ is recorded.

- A graph is plotted with mf and volume of $K_2Cr_2O_7$. The steep rise is end point of the titration.

4. DETERMINATION OF PH BY EMF METHOD:

The pH of a solution could be determined by employing any electrode reversible to H^+ ions. The hydrogen electrode and quinhydrone electrode/ glass electrode are employed for the estimation of pH of a solution.

The emf of a solution depends on the concentration of H^+ ions (or) pH of the solution. A hydrogen electrode containing solution of unknown pH is paired with a standard calomel electrode. The complete cell may be represented as $Pt, H_2(1 \text{ atm}) / H^+ (\text{unknown}) // KCl (\text{salt solution}) / Hg_2Cl_2, Hg$

The emf of the above cell measured by potentiometer and pH of the unknown solution can be calculated as follows.

The emf of the cell is given by

$$E_{\text{cell}} = E_{\text{right}} - E_{\text{left}}$$

$$E_{\text{cell}} = 0.2415 - (-0.0591\text{pH})$$

$$E_{\text{cell}} = 0.2415 + 0.0591\text{pH}$$

ADVANTAGES:

1. Colored solutions, where the use of indicator is impossible are estimated by potentiometric titrations.
2. Poly basic acids can be titrated in steps corresponding to different steps of neutralization.
3. Solutions containing more than one halide can be analyzed in a single titration against Silver Nitrate ($AgNO_3$).

BATTERIES & CELLS:

BATTERIES:

Battery is an electro chemical cell, when two (or) more electro chemical cells are connected in the series, each of which containing two electrodes and an electrolyte is called a battery. It is used a single cell.

i. e. a device which converts a chemical energy to electrical energy is called Battery , a group of two (or)more electrical cells connected together electrically in series ,Batteries are commercial electrochemical cells.

REQUIREMENTS FOR BATTERIES:

1. High capacity, which is small variation of voltage during discharge.
2. High energy efficiency.

$$\% \text{ of efficiency} = \frac{\text{energy released on discharge}}{\text{energy required for charge}} \times 100$$

3. High cycle life is required which is the number of charging and discharging cycles before failure.
4. Long self life is required.

➤ Batteries are classified into 3 types.

1. Primary batteries.
2. Secondary batteries.
3. Fuel cells

Primary batteries	Secondary batteries	Fuel cells
1. Cell reaction is not reversible 2. Can not be recharged. 3. Can be used as long as the materials are active in their composition. 4. Ex: leclanche cell (or) dry cell	1. Cell reaction is can reversible 2. Can be recharged 3. Can be used again and again by recharging the cell. 4. Lead-acid storage cell	1. Energy can be withdrawn indefinitely as long as out side supply of fuel is maintained. 2. Do not store energy. 3. Can not be reused. 4. Oxygen –Hydrogen fuel cell.

PRIMARY BATTERIES:

- The cells in which the cell reaction is not reversible, thus when the reactants have been converted to products, no electricity is produced and the cell becomes dead and after that cannot be used.
- The batteries are used as a source of dc power.

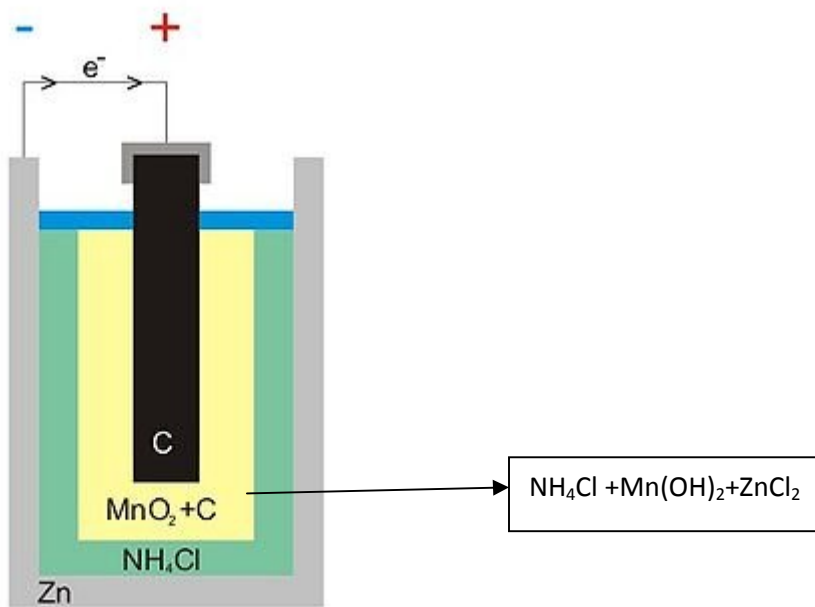
EX: a. LECLANCHE CELL(OR) ZINC – CARBON CELLS (OR) DRY CELL

Dry (or) leclanche cell i.e. a cell without fluid component is a primary cell.

The anode of the cell is zinc container containing an electrolyte NH_4Cl , ZnCl_2 & MnO_2 to which starch is added to make a thick paste like to prevent leakages.

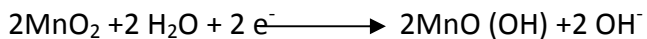
A carbon (graphite) rod acts as a cathode.

CONSTRUCTION:

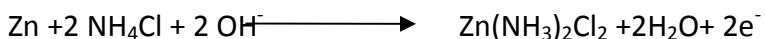


The zinc anode of the cell is shaped into cylindrical can containing the pasty electrolyte into which a carbon (graphite) rod is immersed at the centre.

At the cathode, the reduction reaction occurs with in the moist paste.



A thin zinc cylinder serves as anode and it undergo oxidation.

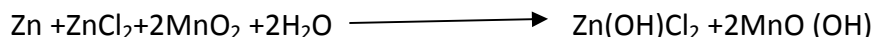


The net reaction is,



ZnCl₂ cells have been found to perform better than the leclanche cells. The electrolyte in a ZnCl₂ cells consists of 15 -40 % Zinc and 60-85 % water, some times with a small amount of NH₄Cl

The overall cell reaction of the zinc chloride as the electrolyte is,



These are very cheap and used widely, but they also have energy density and perform poorly under high current applications.

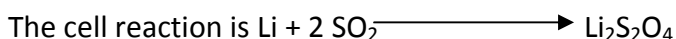
b. LITHIUM AND OTHER METAL CELLS (NON AQUEOUS CELLS)

lithium cells belongs to primary cells. The cells having lithium anodes are lithium cells, irrespective of the cathode used.

Lithium cells can be classified into following.

i. LITHIUM CELLS WITH LIQUID CATHODE:

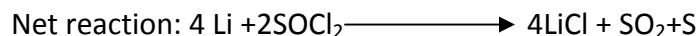
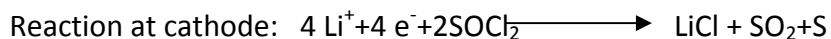
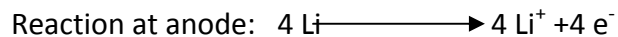
Among Lithium cells with liquid cathode is lithium – sulphur dioxide cell. The co-solvents used are either acrylo Nytrile (or)propylene carbonate (or)mixture of the to with 50% by volume of SO₂



The second type of this category is lithium thionyl chloride cells, which consists of a high surface area carbon cathode.

The thionyl chloride acts as the electrolyte solvent and the active cathode.

The reactions of the cell are:



Due to nature of reaction lithium thionyl chloride cell possess very high energy density. The reaction involves to electrodes per thionyl chloride. Further the SO₂ liberated as product is a liquid under the internal pressure of the cell. No co- solvent is required for the solution as

thionyl chloride is a liquid having moderate vapor pressure, the discharging voltage is 3.3 to 3.5 volts.

APPLICATIONS:

1. Knowing to the excellent voltage control Li – SOCl₂ cells are used on electronic circuit board for supplying fixed voltage for memory protection and other stand by functions.
2. These cells are also used for military space applications.
3. These cells are also used in medical devices such as neuro stimulators, drug -delivery systems.

ii. LITHIUM CELLS WITH SOLID CATHODE:

- The systems may have solid (or) liquid electrolyte. Lithium – manganese dioxide (Li – MnO₂) is emerging as most widely used 3V solid cathode in lithium primary battery.
- The cathode MnO₂ should be heated to >300⁰C to remove water before incorporating it in the cathode.
- It is very important to improve the efficiency of the cells.

Reaction at anode: $\text{Li} \longrightarrow \text{Li}^+ + \text{e}^-$

Reaction at cathode: $\text{Li}^+ + \text{e}^- + \text{MnO}_2 \longrightarrow \text{LiMnO}_2$

Net reaction: $\text{Li} + \text{MnO}_2 \longrightarrow \text{LiMnO}_2$

The electrolyte is a mixture of propylene carbonate.

APPLICATIONS:

1. The cylindrical cells are used in fully automatic cameras.
2. The coin cells are widely used in electronic devices such as calculators and watches.

iii. LITHIUM CELLS WITH SOLID ELECTROLYTE:

- These cells perform best in low current applications and have a very long service life. For this reason, they are used in pace makers.
- Lithium iodine cells used solid Lil as their electrolyte and also produce Lil as the cell discharges.
- The cathode is poly 2 – vinyl pyridine (P2VP)

Reaction at anode: $2\text{Li} \longrightarrow 2\text{Li}^+ + 2\text{e}^-$

Reaction at cathode: $2\text{Li}^+ + 2\text{e}^- + \text{P2VP}_{2n}\text{I}_2 \longrightarrow \text{P2VP}_{(n-1)}\text{I}_2 + 2\text{Lil}$

Net reaction: $2\text{Li} + \text{P2VP}_{2n}\text{I}_2 \longrightarrow \text{P2VP}_{(n-1)}\text{I}_2 + 2\text{Lil}$

APPLICATIONS:

These cells are mainly used in pace makers.

SECONDARY CELLS/STORAGE CELLS:

The cells in which the cell reaction is reversed by passing direct current in opposite direction. These cells are rechargeable and reusable. The secondary batteries can be used through a large number of cycles of discharging.

A combination of all reversible electro chemical cells gives secondary batteries. For example lead acid storage cells, Nickel- Cadmium are widely used as storage cells.

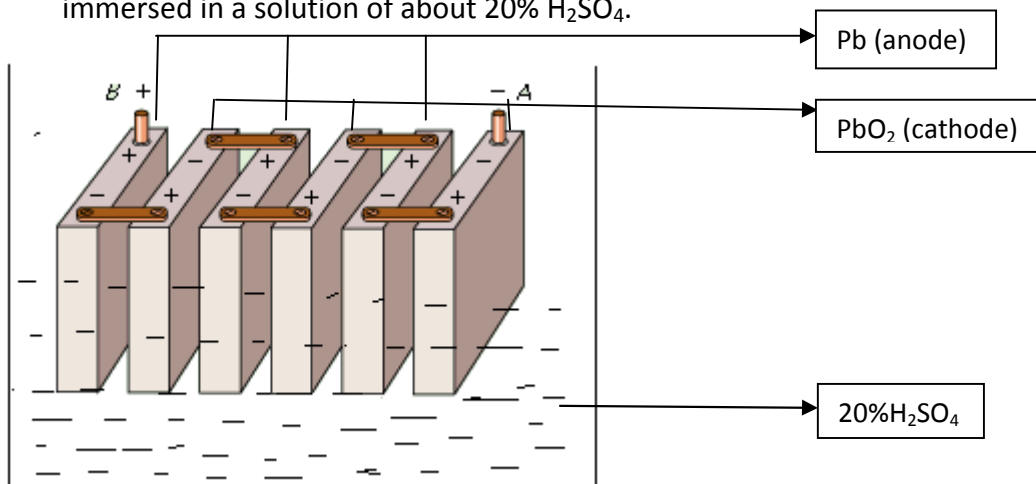
LEAD – ACID STORAGE CELL (LEAD STORAGE BATTERY)

The cell in which original reactants are regenerated by passing direct current from external source, i.e., it is re-charged, is called secondary cell. Lead storage battery is the example of this type.

- a storage cell that can operate both as a voltaic cell and as an electrolytic cell.
- In this cell sponge metallic lead is anode and lead dioxide (PbO_2) is a cathode.

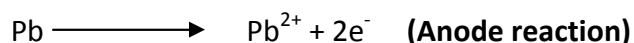
CONSTRUCTION:

- It consists of a group of lead plates (-Ve plates) bearing compressed spongy lead, alternating with a group of lead plates bearing lead dioxide, PbO_2 . These plates are immersed in a solution of about 20% H_2SO_4 .

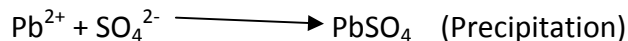


DISCHARGING:

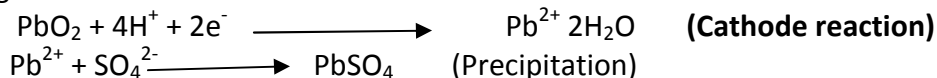
When the storage cell is operating as a voltaic cell (i.e. chemical energy into electrical energy). It is said to be discharging. The lead electrode loses e^- s, which flow through the wire.



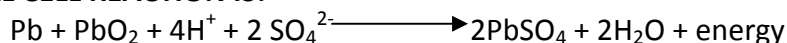
Pb^{2+} ions combine with sulphate ions to form insoluble lead sulphate, PbSO_4 , which begins to coat lead electrode.



The e^- released from the anode (lead plate-Pb) flows to the cathode (lead dioxide- PbO_2). Here PbO_2 gains e^- s to form Pb^{2+} ions.



OVERALL CELL REACTION IS:

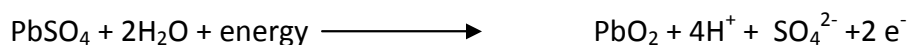


It may be noted that lead sulphate (PbSO_4) is precipitated at both the electrodes. The voltage of each cell is about 2V at a concentration of 21.4% H_2SO_4 at 25°C . lead commonly used in automobiles is a combination of six such cells in series to form a battery with an mf of 12V.

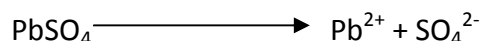
CHARGING:

when both anode and cathode become covered with PbSO_4 the cell ceases to function as a voltaic cell. To recharge a lead storage cell, the reactions taking place during discharging are reversed by passing an external mf greater than 2 volts from a generator. The +Ve pole of the cell (battery).

REACTION AT ANODE:



REACTION AT CATHODE:



NET REACTION DURING CHARGING :



E_{cell} is 2.041 volt.

During the process of charging, the electrodes of the cell are restored to their original conditions (i.e. Pb and PbO_2 respectively). It may also be noted that during discharging operation, the concentration of H_2SO_4 decreases, while concentration of acid increases during the charging operation.

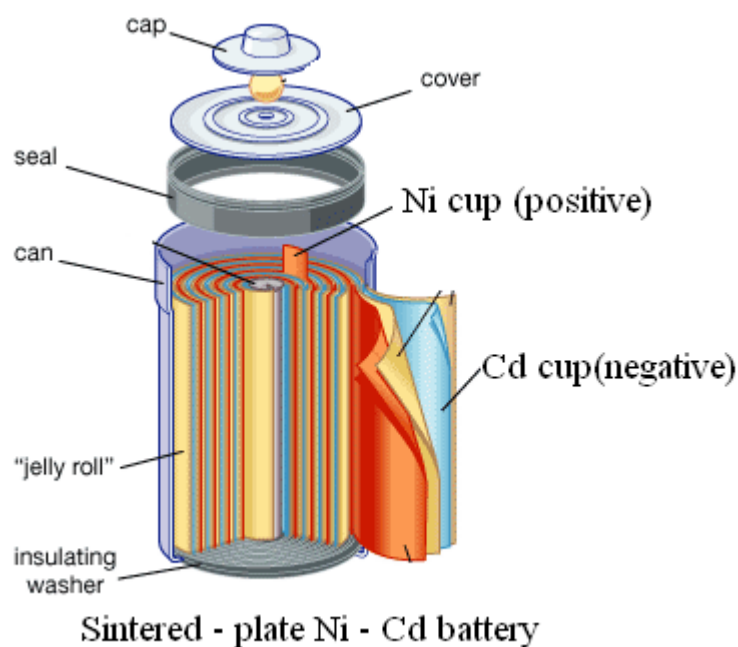
APPLICATIONS OF LEAD – ACID STORAGE BATTERY:

The lead acid storage cells are used to supply current for electrical vehicles, gas engine ignition. In telephone exchanges, electric trains, mines, laboratories, hospitals, broad casting stations, automobiles, power stations.

ADVANTAGES OF LEAD BATTERIES:

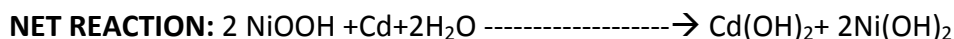
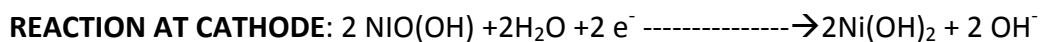
1. The lead acid storage battery is both a voltaic cell and electrolytic cell. When electricity is being drawn from the cell, to start the car it acts as an voltaic cell, when the car is running the cell is being recharged an electrolytic cell.
2. It has relatively constant potential i.e. 12V
3. It is portable and cheap.

b. NICKEL – CADMIUM CELLS:



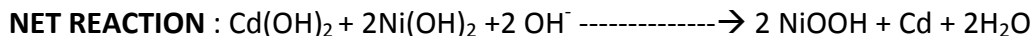
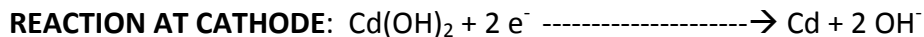
The battery consists of a cadmium anode (-Ve electrode) and a cathode is a composed paste of NiO (OH) ((+Ve electrode). Electrolyte is KOH.

Discharging:



This reaction can be readily reversed, because the reaction products Ni(OH)_2 and Cd(OH)_2 adhere to the electrode surfaces.

CHARGING:



There two types of batteries

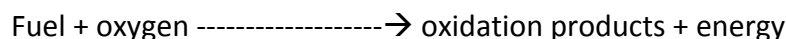
- Pocket plate Ni – Cd battery.
- Sintered plate Ni – Cd battery.
- Ni – Cd battery recently developed is portable, rechargeable cell and its cell voltage is fairly constant (1.4V)
- Like a dry cell it can be packed in a sealed container.

APPLICATIONS:

It is used in electronic calculators, electronic flash units, cordless electronic shavers, transistors and other battery powered small tools.

FUEL CELL:

- Fuel cell is an electrochemical cell which converts chemical energy contained in a fuel oxidation system into electrical energy.
- The only difference between electrochemical cell and fuel cell is that in fuel cell the chemical energy is provided by a fuel and an oxidant stored out side the cell.
- The fuel and oxidizing agents are continuously and separately supplied to the electrodes of the cell at which they undergo reaction.
- Fuel cells are capable of supplying current as long as the reactants are supplied.
- Fuel cells are characterized by
 1. High efficiency
 2. Low noise levels.
 3. Free from vibration, heat transfer and thermal pollution.
 4. Have built in wide range of power requirements.

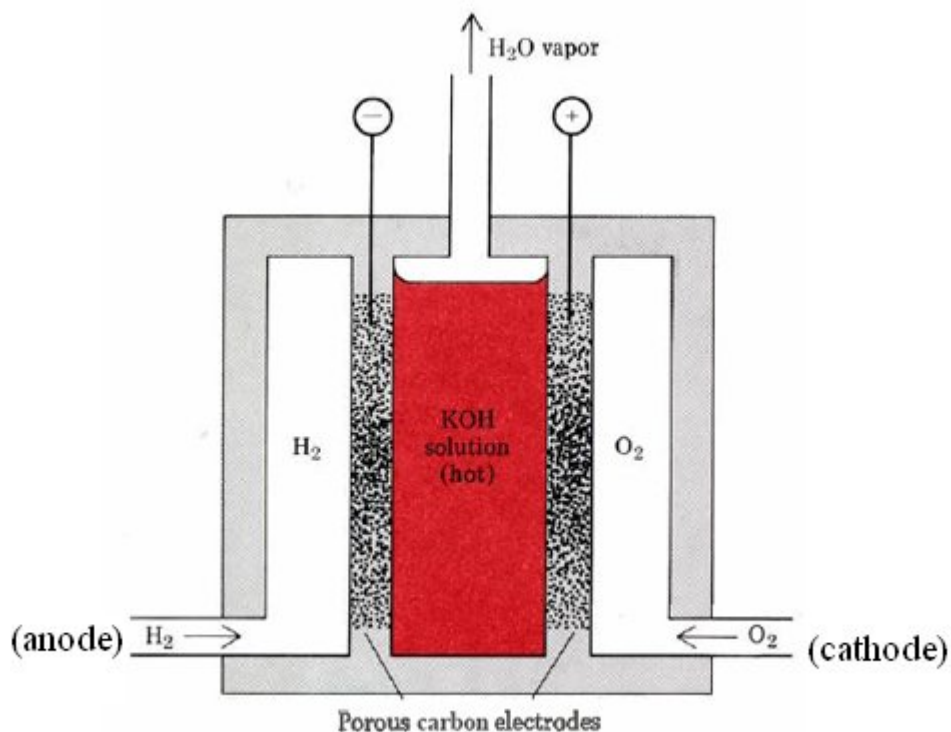


Ex: 1. Hydrogen – oxygen fuel cell ($\text{H}_2\text{-O}_2$).

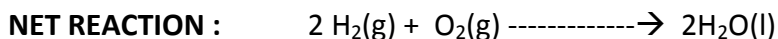
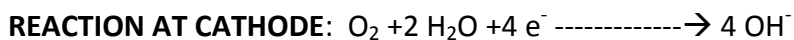
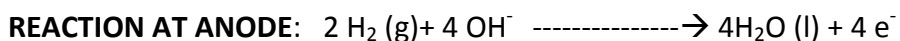
2. Propane – Oxygen fuel cell.

3. Methyl alcohol – oxygen fuel cells.

HYDROGEN – OXYGEN FUEL CELL (H₂-O₂ FUEL CELL):



- The cells consists of two inert porous electrodes (made of impregnated with finely divides Pt (or) a 75/25 alloy of Pb with Ag (or) Ni)
- Electrolytic solution is 2.5 % KOH.
- Through the anode hydrogen gas & through cathode oxygen gas is bubbled.



The products discharged is water and the standard emf of the cell is $E^0 = 1.23\text{V}$. a number of such fuel cells are connected in the series to make ab a battery.

APPLICATIONS:

1. These are used as auxiliary energy source in space vehicles submarines or other military vehicles.
2. Because of light weight these fuel cells are preferred in space craft and product water is available as valuable fresh source for astronauts.

ADVANTAGES:

1. Energy conversion is very high (75 to 82 %).
2. Noise and thermal pollution are less.
3. Maintenance cost is low.

LIMITATIONS:

1. The life time of these fuel cells is not accurately known.
2. Cost is high.
3. Uniform distribution of hydrogen is not possible.

This type of cells are used in space-crafts. Fuel cells are efficient and pollution free.